

Companion XCIX–C

Optimal Probes and Benchmark Suite for RDCC

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Abstract

This merged Companion unifies the content of Companions XCIX–C into a single document. It presents the optimal-probe framework for RDCC inference, including criteria for selecting geometric, morphological, and topological observables, and provides the benchmark suite used to validate these probes across simulations, noise realizations, and tomographic configurations. Together, these components form the operational backbone of RDCC data analysis.

1 Optimal Probes for RDCC Inference

The RDCC framework induces scale-dependent and morphology-dependent signatures in the convergence field. Identifying the optimal probes for extracting these signatures is essential for efficient and robust inference. This section summarizes the criteria for selecting optimal geometric, morphological, and topological observables.

1.1 Definition of an optimal probe

An observable O is considered an *optimal probe* of the relaxation parameter α if it satisfies:

1. **High Fisher information:**

$$g_O(\alpha) = \mathbb{E} \left[(\partial_\alpha \log \mathcal{L}_O)^2 \right] \quad \text{is large;}$$

2. **Low degeneracy:** the observable responds to α in a direction orthogonal to existing probes;
3. **Stability:** the response is robust across noise realizations and tomographic schemes;
4. **Computational efficiency:** the observable can be measured and modeled at scale.

These criteria ensure that optimal probes contribute uniquely and efficiently to the inference pipeline.

1.2 Geometric probes

Geometric probes include:

- convergence power spectra,
- tomographic cross-spectra,

- angular correlation functions.

Their strengths:

- well-understood modeling,
- high signal-to-noise on linear and quasi-linear scales,
- strong constraints on the amplitude and shape of RDCC distortions.

Limitations:

- reduced sensitivity on nonlinear scales,
- partial degeneracy with noise and baryonic effects.

1.3 Morphological probes

Morphological probes include:

- peak counts,
- void statistics,
- Minkowski functionals.

Their strengths:

- strong sensitivity to intermediate-scale distortions,
- complementary degeneracy structure relative to geometric probes,
- robustness to certain noise patterns.

Limitations:

- increased modeling complexity,
- sensitivity to smoothing and filtering choices.

1.4 Topological probes

Topological probes include:

- persistent homology,
- Betti curves,
- transport-based topology.

Their strengths:

- highest sensitivity to nonlinear relaxation effects,
- strong breaking of degeneracies,
- stable information content across noise realizations.

Limitations:

- computationally expensive,
- requires careful treatment of sampling noise.

1.5 Combined optimal-probe strategy

The optimal RDCC probe combination is a hybrid of:

- geometric probes for large-scale structure,
- morphological probes for intermediate-scale sensitivity,
- topological probes for nonlinear relaxation signatures.

This multi-layered structure forms the foundation of the benchmark suite presented in the next section.

2 Benchmark Suite and Validation

To evaluate the performance of optimal probes and to ensure the robustness of RDCC inference, we construct a comprehensive benchmark suite. This suite quantifies the sensitivity, stability, and degeneracy structure of each probe and validates their performance across simulations, noise realizations, and tomographic configurations.

2.1 Structure of the benchmark suite

The benchmark suite consists of:

- **Simulation grid:** A grid of RDCC simulations spanning a range of α values, noise levels, and smoothing scales.
- **Probe library:** A collection of geometric, morphological, and topological observables measured consistently across the simulation grid.
- **Response functions:** Numerical derivatives $\partial_\alpha O$ for each observable O , estimated using finite differences.
- **Fisher matrices:** Probe-specific and combined Fisher matrices used to quantify information content and degeneracy structure.
- **Null-direction analysis:** Identification of weakly informative directions in observable space.

2.2 Validation criteria

Each probe is validated according to:

1. **Sensitivity:** The magnitude and stability of $\partial_\alpha O$ across simulations.
2. **Consistency:** Agreement between independent noise realizations and tomographic schemes.
3. **Degeneracy breaking:** The ability of the probe to break degeneracies present in geometric observables.
4. **Cross-probe alignment:** Stability of the probe’s response direction relative to other probes.
5. **Computational feasibility:** Runtime and memory requirements for large-scale surveys.

2.3 Results of the benchmark suite

The benchmark suite reveals a clear hierarchy:

- **Geometric probes** perform best on large scales and provide stable baseline constraints.
- **Morphological probes** add significant information on intermediate scales and break several geometric degeneracies.
- **Topological probes** provide the strongest constraints on nonlinear relaxation effects and dominate the high-information channels.
- **Combined probes** achieve near-optimal performance, capturing almost all available information about α .

2.4 Implications for RDCC analysis

The benchmark suite provides:

- a validated set of optimal probes,
- a robust framework for multi-probe inference,
- a quantitative understanding of degeneracy structure,
- and a foundation for future survey applications.

This completes the operational backbone of RDCC inference and prepares the ground for the final conclusions of this merged Companion.

3 Conclusion

This merged Companion unifies the optimal-probe framework and the benchmark suite for RDCC inference. The optimal-probe analysis identifies the geometric, morphological, and topological observables that provide the strongest and most complementary constraints on the relaxation parameter α . The benchmark suite validates these probes across simulations, noise realizations, and tomographic configurations, revealing a clear hierarchy of information content and degeneracy-breaking power.

Together, these components form the operational backbone of RDCC data analysis:

- geometric probes provide stable large-scale constraints,
- morphological probes enhance intermediate-scale sensitivity,
- topological probes dominate nonlinear relaxation signatures,
- and combined probes achieve near-optimal performance.

This merged document replaces Companions XCIX–C and provides a unified reference for probe selection, validation, and multi-probe inference in the RDCC framework.

References

- [1] M. Lehmann, *Relaxation-Driven Cyclic Cosmology (RDCC): Flagship Paper*, (2026).
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- [5] M. Lehmann, *RDCC Companion Series: Optimal Probes and Benchmark Suite*, Companions XCIX–C (merged), (2026).